

Vetting TESS Exoplanet Candidates with TRICERATOPS and LEO-Vetter



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Abstract

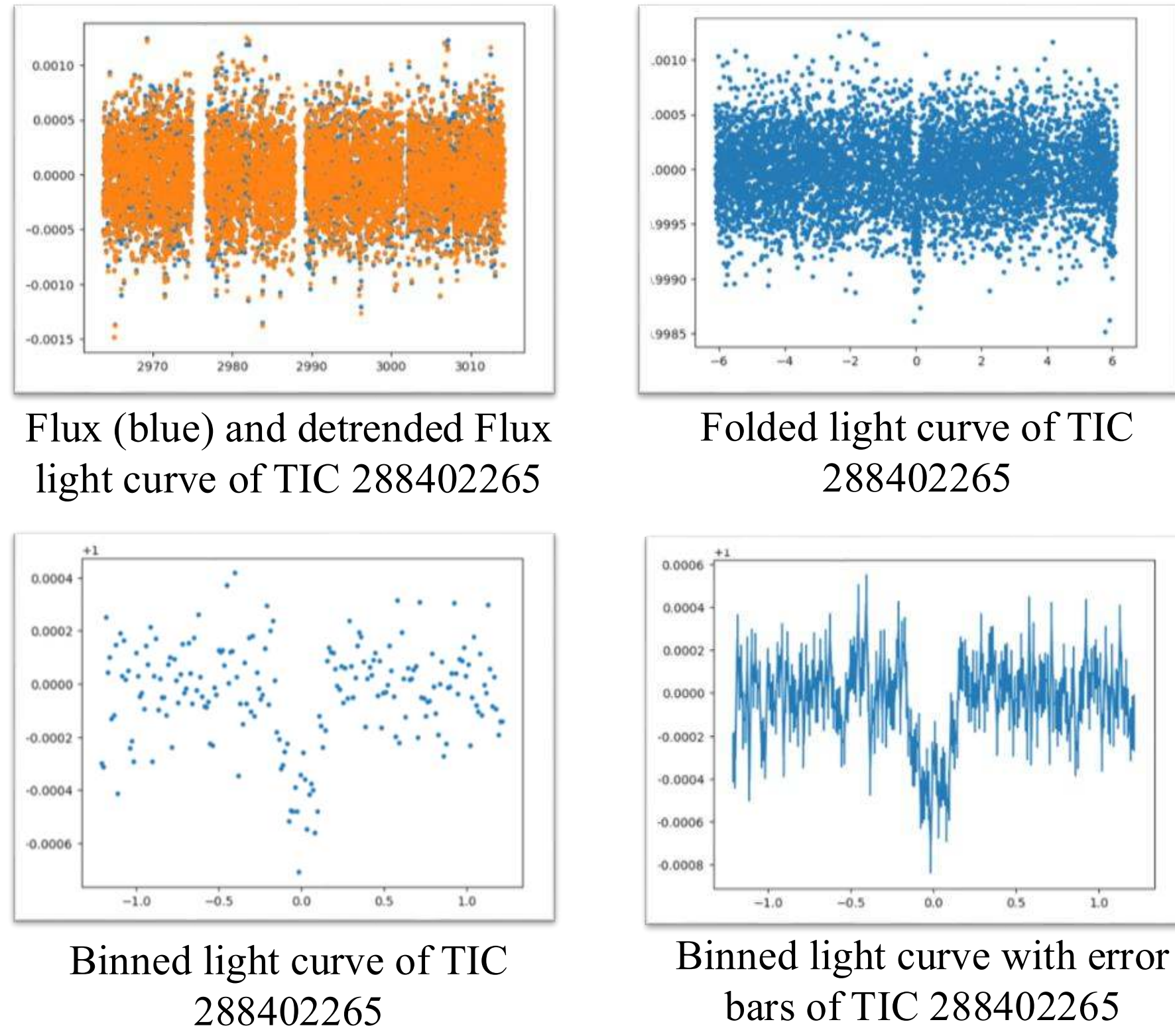
The TESS Science Processing Operation Center searches for transiting planets in 160,000 targets selected from the full frame images (FFI) each observing sector. The pipeline detects several thousand potential transit signals, or threshold-crossing events (TCEs) from these targets. The TCEs must be vetted to see if they are false positives or planet candidates. We vetted the TCEs from the FFI multi-sector search of sectors 56-69. We first selected TCEs that had ExoMiner machine-learning classification scores of greater than 0.5 (likely planet candidate) and then vetted them using the TRICERATOPS and LEO-Vetter vetting tools. We identified 9 objects of interest with radii between 1.26 and 8.19 $R_{\oplus}$ and orbital periods between 1.22 and 36.91 days. We performed light curve modeling using the Allessfitter package to get the best-fit transit parameters for these nine targets.

Introduction

The Transiting Exoplanet Survey Satellite (TESS) continues its mission of searching for transiting exoplanets (Ricker et al. 2014). The TESS Science Processing Operations Center (SPOC) pipeline creates light curves for around 160,000 targets each observing sector from the full frame images (FFIs; Caldwell et al. 2020). A search for transiting planets on these 160,000 targets in any given sector finds around 3,000 to 5,000 threshold-crossing events (TCEs), though only a small number of these TCEs are good planet candidates. These TCEs need to be further examined to determine if they are valid planet candidates or false positives. As a first filter on these TCEs, we run ExoMiner, a deep learning algorithm used for vetting TESS and Kepler planet candidates (Valizadegan et al. 2022, 2025). All of the TCEs are run through ExoMiner, which makes predictions on the TCEs being planet candidates, resulting in a mean score between zero (high-confidence false positive) and one (high-confidence planet candidate). We then use other vetting tools to determine which ExoMiner TCEs from sectors 56-69 are likely to be planet candidates. The first, TRICERATOPS, is a Bayesian tool used for vetting and validating TESS Objects of Interest (Giacalone et al. 2021). TRICERATOPS identifies neighboring stars bright enough to create a transit as seen in the light curve of a TCE. It then models different planet and false-positive scenarios and determines the probabilities of each scenario. The second tool, LEO-Vetter, runs a series of pass-fail thresholds to vet the light curve data (Kunimoto 2024). It checks for astrophysical false positives and for false alarms caused by instrumental or observing

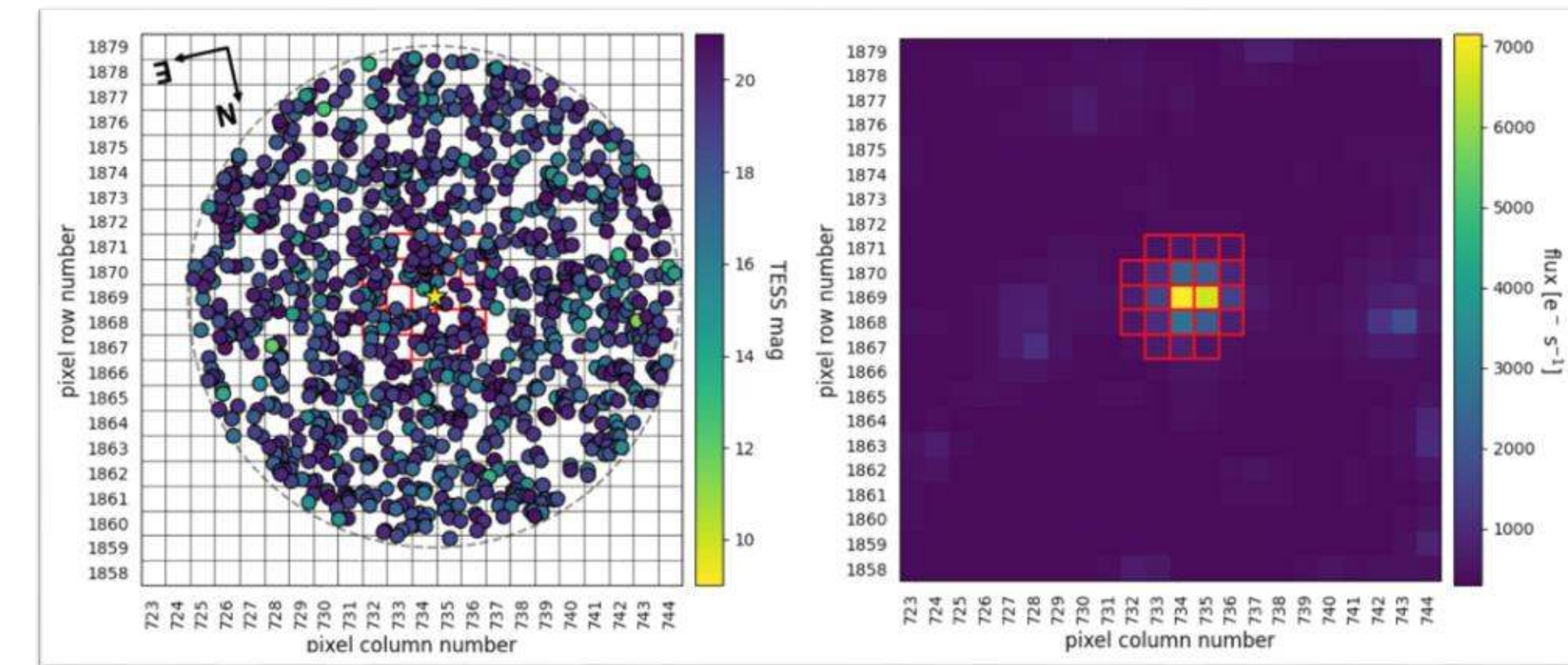
Methods

We first obtained the Data Validation (DV) Time Series files for the TCEs with ExoMiner scores above 0.5 that were observed in multiple sectors between Sectors 56 and 69 and were not already known TESS objects of interest (TOI) from the TOI catalog. We used the SPOC Data Validation (DV) time series Detrended Flux light curves as input to TRICERATOPS. We folded the light curve on the TCE orbital period found by the SPOC pipeline and binned the light curve prior to input to TRICERATOPS. We made some minor changes to the TRICERATOPS code for it to work with the TESS FFI data, rather than the 2-minute cadence data. We then ran the TCEs through TRICERATOPS with twenty trails for each TCE. We recorded the mean false positive probability (FPP) and nearby false positive probably (NFPP) for each TCE as well as the probabilities for each scenario modeled by TRICERATOPS. We ran 27 TCEs with low NFPP and FPP through LEO-Vetter which checked for false alarms and false positives. The TCEs that had low probabilities of NFPP and FPP and passed the LEO-Vetter false alarm and false positive test were labeled as planet candidates. Some planet candidates had NFPP and FPP probabilities that were greater than the TRICERATOPS planet candidate limit of $\text{NFPP} < 10^{-3}$ and $\text{FPP} < 0.5$. We used the DV difference image offset limits to rule out stars contributing to the TRICERATOPS NFPP that were unlikely to be the cause of the transit signal; see, e.g. Figure 1. Stars that contributed to the TRICERATOP NFPP but were outside the 3σ uncertainty radius of the DV determined location of the transit signal were excluded from the NFPP calculation.



FPP = 0.07080156554088465 +/- 0.012585582883994372
NFPP = 0.03788584597922894 +/- 0.0032456201400017292

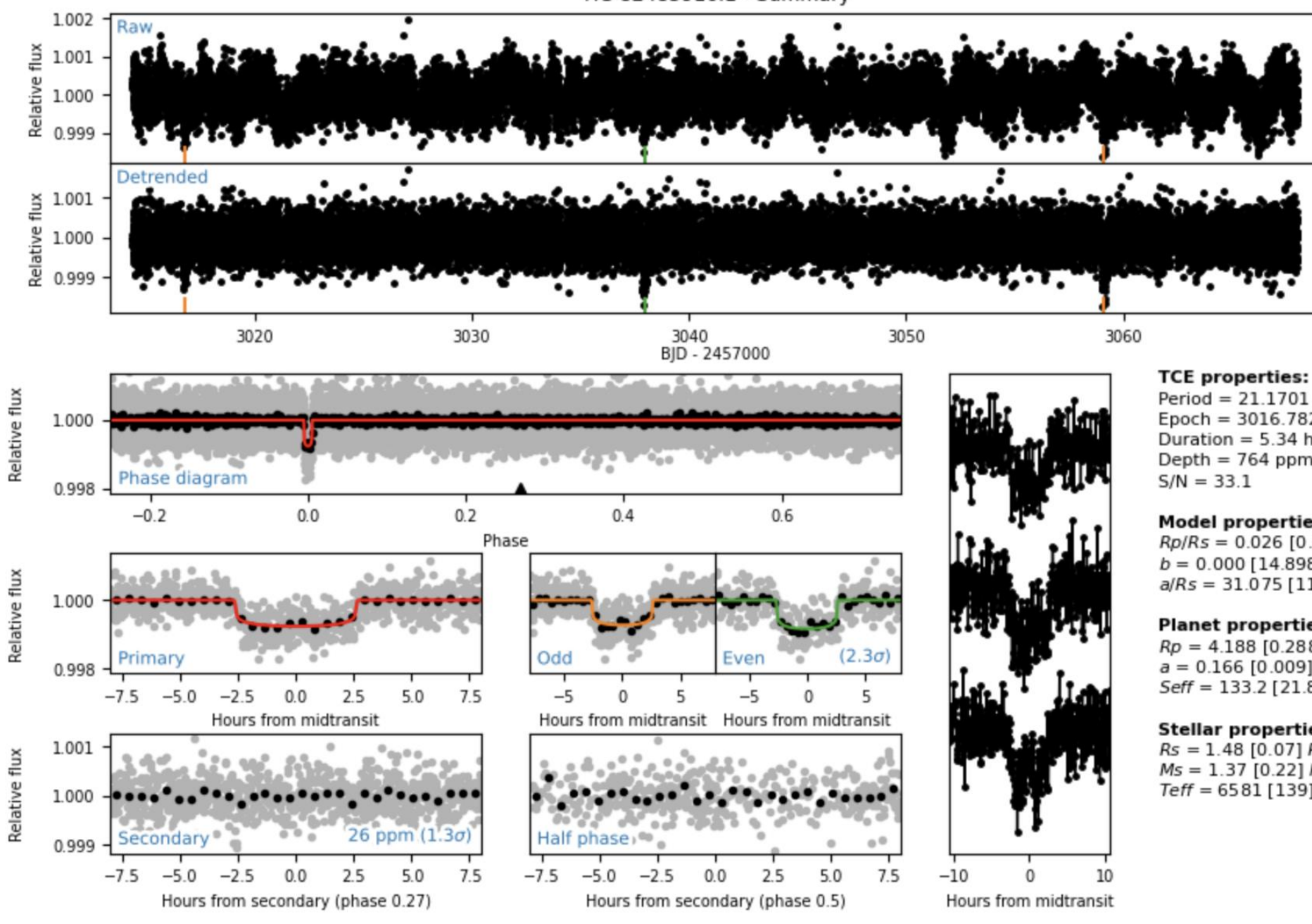
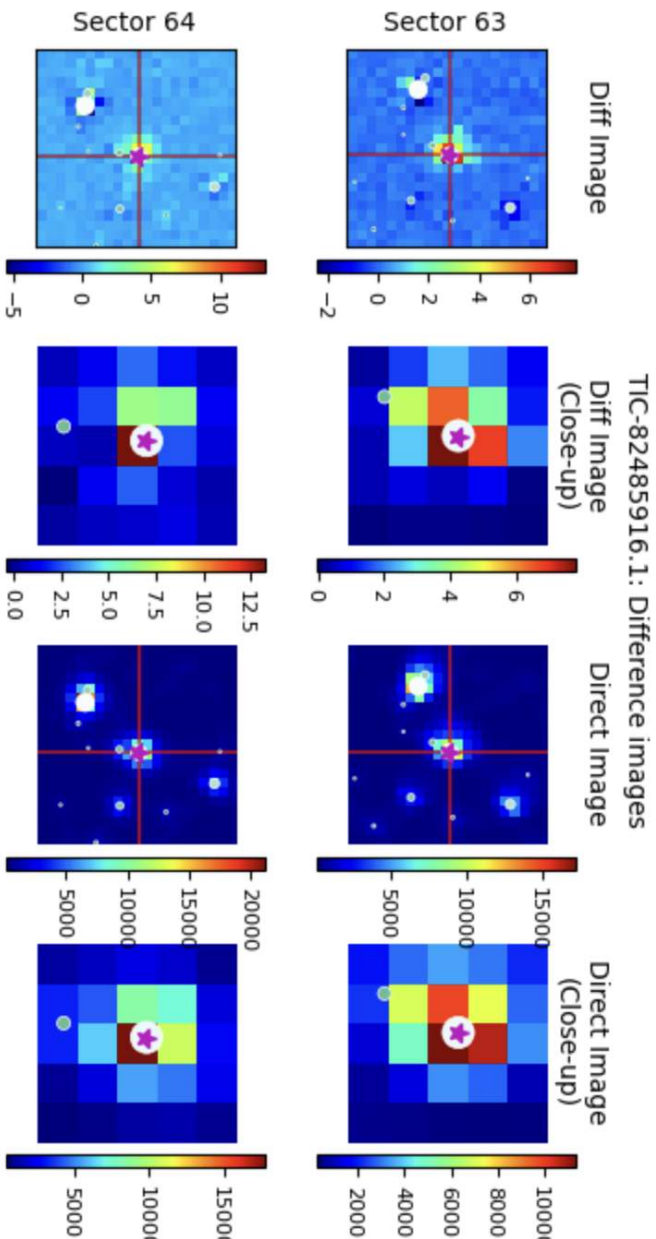
The image on the left shows the nearby stars and their magnitude that TESS views as well as the pixels utilized to form the light curve. The image on the right shows the difference image of the target star as well as the apertures used to form the light curve.



False Positive Probability and Nearby False Positive Probability of TIC 388402265

	ID	scenario	M_s	R_s	P_orb	inc	b	ecc	w	R_p	M_EB	R_EB	prob
0	388402265	TP	1.950000	2.550780	12.232634	86.769626	0.578853	0.179443	170.065883	5.730789	0.000000	0.000000	0.708902
3	388402265	PTP	1.950000	2.550780	12.232634	86.660786	0.445713	0.408073	151.950174	6.188461	0.000000	0.000000	0.151523
6	388402265	STP	1.023507	1.113501	12.232634	89.972757	0.009167	0.437542	200.321769	9.092936	0.000000	0.000000	0.024579
7	388402265	SEB	0.605749	0.598128	12.232634	89.895072	0.034911	0.662462	180.175560	0.000000	0.141984	0.172887	0.009157
8	388402265	SEBx2P	0.383259	0.387874	24.465269	89.623424	0.272245	0.670879	171.126338	0.000000	0.367944	0.374985	0.000132
9	388402265	DTP	1.950000	2.550780	12.232634	86.754248	0.450975	0.280833	109.037803	5.521238	0.000000	0.000000	0.056994
12	388402265	BTP	1.228000	2.312773	12.232634	89.352979	0.135088	0.291907	314.292621	16.603365	0.000000	0.000000	0.000492
13	388402265	BEB	0.965000	1.101026	12.232634	89.326768	0.260090	0.488408	210.638971	0.000000	0.280833	0.301456	0.007383
14	388402265	BEBx2P	0.865000	1.226145	24.465269	88.865975	0.915181	0.490949	242.323931	0.000000	0.822729	0.849098	0.000485
16	817497946	NEB	1.000000	1.000000	12.232634	89.955381	0.018001	0.607271	210.844803	0.000000	0.438126	0.436868	0.001079
17	817497946	NEBx2P	1.000000	1.000000	24.465269	88.986106	0.671122	0.250534	157.503709	0.000000	0.955110	1.000000	0.000244
19	388402262	NEB	1.450000	1.420300	12.232634	88.566233	0.293806	0.514029	158.937925	0.000000	0.286455	0.306286	0.009973
20	388402262	NEBx2P	1.450000	1.420300	24.465269	87.940131	0.872514	0.390417	142.207187	0.000000	1.421563	1.420300	0.000572

Difference Images of TIC 82485916 in LEO-Vetter for sectors 63 and 64



LEO-Vetter Summary for TIC 82485916 with raw and detrended light curves. LEO-Vetter checked for false alarms and false positives including: Inconsistent Transit SNR, Unphysical Transit Orbit, Odd-Even Transit Difference, Not Enough Valid Transits, Unique Events, Planet Radius To Large, Significant Secondary, Off Target

TIC 370456180 difference image offset limits from the TESS-SPOC pipeline Data Validation test. The multi-sector average offset of the transit signal position from the TIC position (red asterisk) is shown with a red cross. The circle represents the 3σ uncertainty for the position. Locations for the measured transit signal positions in sectors 56 and 57 are indicated by green crosses. Additional catalog star locations are indicated by blue asterisks and annotated with their TIC ID and magnitude.

Planet Candidates

We identified nine planet candidates in the multi-sector search of sectors 56-69. Three are Super-Earths with radii between 1.26 and 1.94 R_E and orbital periods between 1.22 and 2.61 days. One is a sub-Saturn with a radius of 8.19 R_E and an orbital period of 5.4 days, and five are Neptunian planets with radii between 3.66 and 4.83 R_E and orbital periods between 6.8 and 36.91 days. The planet candidates were modeled with Allessfitter (Günther & Daylan 2021) to get fitted planet parameters. It is notable that no ExoMiner planet candidates with $R_p > 10R_{\oplus}$ passed our vetting tests.

Table 1. Planet Candidates and their DV parameters with TRICERATOPS NFPP and FPP													
TIC	Period	Epoch	Depth (ppm)	Rp(R _s)	Teq (K)	Tmag	NFPP	FPP	ExoMiner Score	RUWE			
294668702	36.90633	2979.8662	1441.41	4.83	600	12.17	0.005600	0.01716944746	0.798114004	1.015			
370456180	6.81861	2830.8271	463.71	4.63	1360	10.15	0.01400	0.02997753389	0.628358316	0.973			
348899056	2.16702	2964.179	710.6	1.68	817	11.69	0.006874	0.1183518782	0.794543722	1.105			
288769849	1.22933	3069.7697	444.6	1.26	1026	10.22	0.032374	0.1348155704	0.78768529	0.950			
257711889	5.3987	3156.1244	2617.4	8.19	1183	12.53	2.695 × 10 ⁻⁴	0.008809655845	0.824648348	0.822			
410399327	2.60811	3099.3009	442.5	1.94	1224	10.47	0.003341	0.05436296946	0.838800922	1.03			
82485916	21.1698	3016.7827	716.5	4.41	866	8.25	2.422 × 10 ⁻¹⁰	0.01503735576	0.800580364	0.978			
353855214	23.17212	2842.194	869.5	3.66	713	11.39	3.046 × 10 ⁻⁹	0.007917079529	0.834226762	0.937			
29778632	25.62358	2966.515	956.6	4.32	716	11.69	0	0.006071749641	0.83286645	0.966			

Conclusion

We utilized ExoMiner, TRICERATOPS and LEO-Vetter vetting tools to identify nine planet candidates. The planet candidates will be submitted to the Exoplanet Follow-up Observing Program (ExoFOP) as Community TESS Objects of Interest to be further vetted. With TRICERTOPS we were able to rule out known stars near our targets, but we were not able to rule out faint nearby stars that were not in the catalogs used by TRICERATOPS. Faint nearby stars will have to be ruled out by further observations from ground-based telescopes.

Future Work

The future step of this project is to run these tests on future TCE targets from the TESS SPOC Pipeline. An updated version of ExoMiner fine-tuned for use on TESS TCEs is also being created, which once tested could negate the need to use TRICERATOPS and LEO-Vetter for vetting TESS TCEs.

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